

Visualizing sound with microphone array and mixed reality headset

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Proposal

Motivation

Sound is an important sensory indicator of one's environment. This project aims to visualize the location and type of surrounding sounds to help mitigate the impact of situational, temporary and permanent impairments. This visualization of sound has the potential to improve safety in environments such as manufacturing where identifying warning signals is critical.

Aims

This project aims to develop a system that localizes and classifies sound, presented in multiple visualizations to the user to improve awareness and help mitigate possible hearing impairments. The effectiveness of the system and its visualizations will be evaluated in terms of localization accuracy, workload, comfort and perceived usefulness.

Progress

- Tech stack chosen: Python for the sound localization scripting on the microphone array which streams results to Unity scene using a Python WebSocket server.
- Sound localization script working using on-board beamforming to sense the location of one sound source.
- Sound classification is integrated in parallel to sound localization, using Google's MediaPipe Audio Classifier.
- Angles and classifications from sound localization script successfully networked to Unity scene using the Python WebSocket server.
- Informed design by researching possible use cases such as manufacturing environments.
- Pilot Experimental design drafted.

Problems and risks

Problems

- N-way beamforming to localize multiple sound sources required in depth beamforming knowledge; using the on-board 1-way beamforming instead.

- The on-board beam-forming algorithm is hardcoded and not adjustable.

Risks

- It may be difficult to gather participants with experience with manufacturing settings. Mitigation: use simulated scenarios to better represent use cases.
- Visualizations may all garner similar results. Mitigation: will create challenging test environments to better differentiate their effectiveness.

Plan

Semester 2

- Before Week 1 – Research relevant wayfinding literature to inform visualizations chosen to implement.
- Week 1 – 2: Finalize prototype visualizations and Unity integration
 - Deliverable: Have complete visualizations within unity scene that can be used for pilot study and built onto Meta Quest 3 headset.
- Week 3: Run Pilot study
 - Deliverable: Quantitative measures of systems usability and effectiveness for 1 visualization or more. Qualitative data surrounding possible use cases and improvements to inform full pilot study. Study ran with at least 6 participants.
- Week 4: Prepare experimental environment
 - Deliverable: Have fully working unity environments with visualizations implemented representing uncertainty found in microphone array.
- Week 5 – 6: Conduct full user study in simulated environments using a simulation of the sound localization
 - Deliverable: quantitative measures of visualizations usability and effectiveness, qualitative measures of perceived workload, comfort and usefulness. Study ran with at least 10 participants.
- Week 7: Analyze quantitative and qualitative data gathered from study
 - Deliverable: Analysis of visualizations comparing their effectiveness identifying strengths and weaknesses. Suggest improvements and possible modifications for future work based off found evidence.
- Week 8-10: Write up of dissertation
 - Deliverable: first draft of dissertation submitted to supervisor within two weeks before the final submission date.